# Lithium-ion Lifecycle

**Standard Operating Procedure**

**Nicholas Odeh 8/2026**

# Contents

# Purpose

Lithium-ion cells must be stored based on their lifecycle classification. This document outlines the storage, preparation, and packaging requirements for Lithium-Ion batteries, with a focus on Devices identified as DDR (Defective, Damaged, or Recalled) .

**Notice:** Personnel should review the Lithium-Ion Event Response SOP in tandem with the Lithium-Ion Lifecycle SOP.

# What are DDR cells?

DDR cells are batteries that may present an elevated safety risk due to damage, defects, or recall status, including the potential for short-circuiting, overheating, thermal runaway, fire, or rapid combustion.

# Lithium-Ion Lifecycle Management

## Battery Charge Requirements

Cells should be reduced to ≤ 15% SOC (State of Charge) prior to packaging, when feasible.

## Storage Environment Requirements

DDR batteries awaiting disposition or disposal should be stored under controlled environmental conditions that minimize the risk of damage, short-circuiting, overheating, or other battery-related incidents.

- Temperature: 50°F - 65°F
- Humidity: 10% - 40%
- Store batteries on non-conductive shelving or pallets.
- Do not store batteries near fire exits, emergency egress routes, or evacuation pathways.
    i. Review the Lithium Incident Response SOP for further fire readiness guidance.
- Ensure the storage room contains proper fire suppression equipment.

### Storage Location

Storage locations should maintain physical separation between known-good cells and DDR cells to prevent accidental mixing, handling, or reuse.

Access to designated DDR battery storage areas should be restricted to trained and authorized personnel. Only personnel who have received the appropriate safety and handling training may move, package, inspect, or otherwise handle DDR batteries.

The designated DDR battery disposal storage area is located at ***XYZ LOCATON***.

## Preparing DDR cells for disposal

### Locate the battery for disposal

- Identify the specific battery you intend to package.
- Handle it carefully and avoid bending, puncturing, or crushing it.

### Detach Accessories

- Detach any cables, screws, brackets, or external hardware that are not permanently fixed to the battery.
- Do not attempt to remove components that require prying, cutting, puncturing, or otherwise disturbing the battery enclosure.

### Protect Terminals

- Apply polyimide tape to the battery terminals to prevent short circuits.
- Verify that conductive surfaces are fully covered before proceeding with packaging.

### Packaging and Secure Battery

- Place each battery in its own anti-static bag.
- Fold the anti-static bag around the battery.
- Seal the bag using polyimide.
- Do not package multiple batteries within the same anti-static bag.

## Utilizing The Disposal Container

- Prepare an approved non-combustible metal disposal container with a minimum 10-inch base layer of vermiculite.
- Place the individually packaged battery onto the vermiculite base layer.
- Cover the battery with vermiculite so it is fully surrounded with a minimum 1-inch layer on all sides, ensuring it is well cushioned and cannot shift, collide, or be damaged during movement.

## Seal Disposal Container

- Securely seal the metal container and label it per disposal requirements.
- Repeat the above steps until container is full, or until barrel is set for pickup.
- Do not leave DDR batteries in barrel for more than 6 months at a time.

# DDR Checklist

☐ Chemistry identified (Li-ion).

☐ Status identified (DDR).

☐ Battery charge at 15% or less, if possible.

☐ All screws and cables removed from battery cell, if applicable.

☐ Terminals protected with Polyimide tape.

☐ Battery packaged in anti-static bags.

☐ Packaged battery placed in a metal barrel containing at least 10 inches of vermiculite.

☐ Packaged battery is fully cushioned, and cannot reasonably shift, collide, or become damaged during movement.

☐ Stored in a cool, dry environment, and away from ignition sources.

☐ Designated personnel disposed batteries per local and federal regulations.